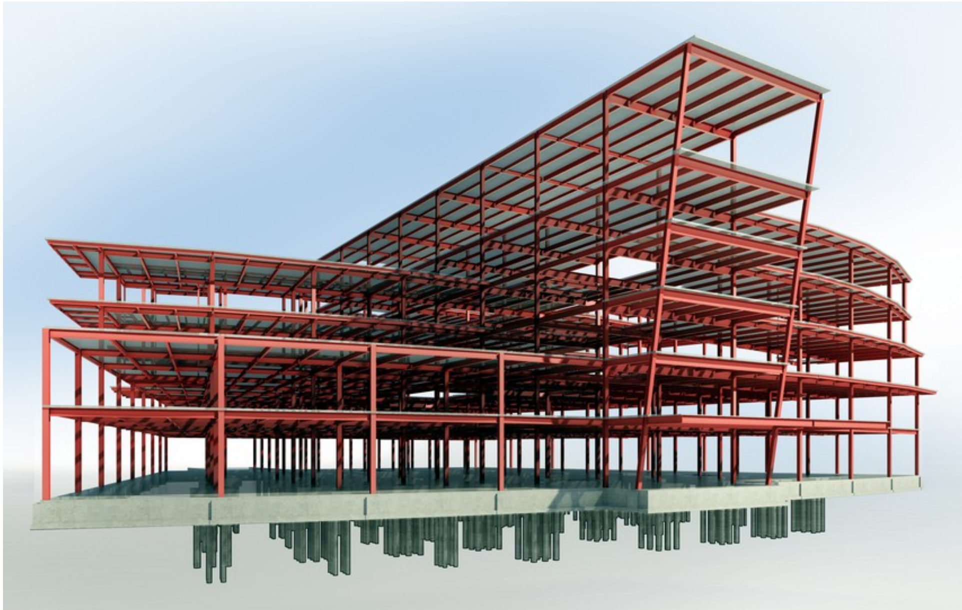
STRUCTRA TRAINING

REVIT FOR STRUCTURES

COURSE OUTLINE — STEEL & CONCRETE STRUCTURAL MODELING IN AUTODESK REVIT

REINFORCED CONCRETE TRACK | STEEL TRACK



Structural BIM model — Autodesk Revit

Program Overview

Revit for Structures is a hands-on Autodesk Revit training program for engineering students and early-career professionals who want to build real, buildable structural models — not just pretty renders. The program runs as two focused tracks that can be taken separately or together: Reinforced Concrete Structures and Steel Structures. Both tracks move from the fundamentals of the Revit interface through full structural modeling, reinforcement/connection detailing, schedules, and construction documentation.

Start date	Monday, 14 September 2026 (third week of September)
Format	100% online — live instructor-led sessions via Zoom
Duration	4 weeks per track · weekday evenings
Fee	KSH 2,500 per track
Who it's for	University engineering students and early-career structural/BIM professionals
Prerequisites	A laptop capable of running Revit; no prior Revit experience required

Reinforced Concrete Structures Track

1. Brief overview of Revit

- Difference with AutoCAD software
- Benefits over AutoCAD software

2. Setting up the software

- Brief introduction of libraries required
- Downloading other extensions from the Autodesk site

3. Beginning with Revit

- Creating a new project
- An overview of the templates
- An overview of palettes, tabs and panels

4. Creating datum elements

- Creating levels
- Creating grids
- Propagating grid extents
- Customizing grid bubble shape and appearance
- Linking CAD files to create grids

5. Overview of creating 3D elements

- Foundations, columns, beams and slabs

6. Creating columns

- Placement of columns
- Loading column families

7. Creating foundations

- Hosting foundations on the columns created
- Creating strip footing
- Using the eccentricity option
- Basement floor slab
- Loading foundation families into your drawings

8. Creating beams and slabs

- Copying beams and slabs from level to level
- Creating a slab
- Hollow pot and waffle slabs

9. Curved beams

- Procedure for modeling curved beams

10. Beam system layout and justification rules

- Using the beam system to model structural beams
- Modeling beams using beam system layout rules
- Modeling beams using beam system justification rules
- Deselecting the beam system

11. Creating staircases and ramps

- Types of staircases
- Creating an architectural ramp and a structural ramp
- Modeling a staircase as a ramp
- Shaft openings

12. Visibility/graphics and filters

- Applying visibility/graphic overrides — method 1
- Applying visibility/graphic overrides — method 2
- Applying visibility/graphic overrides — method 3
- Applying visibility/graphic overrides — method 4

13. Creating roofs and truss placement

- Sloped and gable-ended roofing
- Adding system trusses to your roof
- Editing and creating customized trusses

14. Placing reinforcement

- Creating sections in your drawing
- Foundation reinforcement
- Column reinforcement
- Beam reinforcement
- Slab reinforcement
- Staircase reinforcement
- Ramp reinforcement

15. Annotations

- Adding dimensions
- Adding spot elevations
- Adding spot coordinates
- Adding spot slope

16. Detailing of sections

- Placing breaklines in sections
- Using hatch files to create regions
- Placement of rebar tags on reinforcement
- Adding comments on reinforcement
- Tagging structural elements

17. Schedules and material takeoffs

- Creating bar bending schedules
- Loading shape images into bar bending schedules
- Creating material takeoffs for structural elements
- Sorting and filtering fields in schedules

18. Placing drawings on sheets

- Loading different paper sizes for title blocks
- Using system-loaded title blocks
- Editing and customizing title blocks
- Adjusting text and fields in a title block
- Saving title blocks as a Revit family template

- Duplicate options for drawings
- Placing drawings on sheets

19. Printing sheets to PDF format

- Setting up sheets before printing
- Printing one sheet at a time
- Printing several sheets at a time

20. Converting files into CAD and other formats

- Export options for your drawing
- Transporting your drawing to AutoCAD
- Sending your model for analysis

Steel Structures Track

1. Brief overview of Revit

- Difference with CAD software
- Benefits over CAD software

2. Setting up the software

- Brief introduction of libraries required
- Downloading other extensions from the Autodesk site

3. Beginning with Revit

- Creating a new project
- An overview of the templates
- An overview of palettes, tabs and panels

4. Creating datum elements

- Creating levels
- Creating grids
- Elevation markers
- Propagating grid extents
- Customizing grid bubble shape and appearance (optional)
- Linking CAD files to create grids

5. Modeling steel columns

- Placement of columns
- Loading column families

6. Modeling concrete columns

- Placement of concrete columns
- Loading column families

7. Modeling the foundations

- Hosting foundations on the columns created
- Loading other foundation families into your drawings

8. Modeling the rafters

- Placement of rafters
- Loading steel beam sections
- Attaching columns to the rafters
- Copying rafters from grid to grid

9. Modeling the steel beams

- Loading steel beam sections
- Modeling the floor beams
- Justification rule of the beams
- Copying the floor beams
- Modeling beams at the eaves level

10. Beam system layout and justification rules

- Using the beam system to model structural beams
- Modeling beams using beam system layout rules

22. Connection types

- Main types of connection
- Haunch connection
- Apex connection
- Base plate connection
- Gusset plate bracing connection
- Gusset plate column-base plate connection
- Purlin / side rail connection
- Single-sided end plate (simple connection)
- Single-sided end plate (moment connection)
- Double-sided end plate (simple connection)
- Fin plate (simple connection)
- Clip angle (simple connection)
- Column splice connection
- Creating custom connections

23. Fabrication elements, modifiers and parametric cuts

- Use and location

24. Annotations

- Adding dimensions
- Adding spot elevations
- Adding spot coordinates
- Adding spot slope
- Creating custom bolt and plate tags

25. Detailing of sections

- Placing breaklines in sections
- Using hatch files to create regions
- Placement of rebar tags on reinforcement
- Adding comments on reinforcement
- Tagging structural elements

26. Schedules and material takeoffs

- Creating bar bending schedules
- Loading shape images into bar bending schedules
- Creating material takeoffs for structural elements
- Sorting and filtering fields in schedules

27. Placing drawings on sheets

- Loading different paper sizes for title blocks
- Using system-loaded title blocks
- Editing and customizing title blocks
- Adjusting text and fields in a title block
- Saving title blocks as a Revit family template
- Duplicate options for drawings
- Placing drawings on sheets

28. Printing sheets to PDF format

- Setting up sheets before printing
- Printing one sheet at a time

- Printing several sheets at a time

29. Converting files into CAD and other formats

- Export options for your drawing
- Transporting your drawing to AutoCAD
- Sending your model for analysis